

Week 2 - Command-Line Administrative Guide

This guide explains basic Linux administration commands used in Ubuntu systems. These commands are useful for IT support, software administration, system maintenance, and development tasks used at Ubuntu Innovations (Pty) Ltd. The company uses tools such as Python, PostgreSQL, GitHub, Wireshark, and Windows/Ubuntu administration software from the software inventory.

Command: pwd

Purpose: Displays the current working directory in the Linux file system.

Example: pwd

Why It's Useful: Helps users know exactly where they are working in the system.

Simple Explanation: It shows the folder you are currently inside.

Command: ls

Purpose: Lists files and folders in the current directory.

Example: ls

Why It's Useful: Allows administrators and developers to view files, scripts, and software folders quickly.

Simple Explanation: It shows what files and folders are in the current location.

Command: cd

Purpose: Changes from one directory to another.

Example: cd Documents

Why It's Useful: Helps users move around the Linux file system to access software files, logs, or projects.

Simple Explanation: It lets you move into another folder.

Command: mkdir

Purpose: Creates a new directory (folder).

Example: mkdir Projects

Why It's Useful: Useful for organising software projects, backups, and administrative files.

Simple Explanation: It creates a new folder.

Command: touch

Purpose: Creates a new empty file.

Example: touch notes.txt

Why It's Useful: Useful for creating configuration files, scripts, or documentation files.

Simple Explanation: It creates a new blank file.

Command: chmod

Purpose: Changes file or folder permissions.

Example: chmod 755 script.sh

Why It's Useful: Controls who can read, write, or execute files. This improves system security.

Simple Explanation: It changes what users are allowed to do with a file.

Command: chown

Purpose: Changes the owner of a file or directory.

Example: sudo chown laeeq:laeeq report.txt

Why It's Useful: Helps administrators manage file ownership and user access.

Simple Explanation: It changes who owns the file.

Command: ps

Purpose: Displays currently running processes.

Example: ps

Why It's Useful: Helps administrators monitor active programs and troubleshoot problems.

Simple Explanation: It shows which programs are currently running.

Command: top

Purpose: Shows real-time system processes and resource usage.

Example: top

Why It's Useful: Useful for monitoring CPU usage, memory usage, and system performance.

Simple Explanation: It gives a live view of what the computer is doing.

Command: apt install

Purpose: Installs software packages on Ubuntu Linux.

Example: sudo apt install python3

Why It's Useful: Allows administrators to install software tools such as Python, PostgreSQL, Wireshark, and other applications used by Ubuntu Innovations.

Simple Explanation: It downloads and installs software on Ubuntu.

Conclusion: These Linux commands are important for beginner system administrators, developers, and IT support staff. They help users manage files, monitor systems, install software, and maintain secure and organised computer systems. Many of these commands support the software and administration tasks required at Ubuntu Innovations (Pty) Ltd.